

7.2 Systems of Linear Equations from the Linear-Combination Perspective



Core skills you'll master in this section:

- ▶ The third perspective on matrix multiplication: the result of the matrix multiplication $A\mathbf{x}$ is a linear combination of A 's column vectors.
- ▶ How identity matrices, diagonal matrices, and triangular matrices affect the basis and the shape of the grid.
- ▶ When the determinant is zero: the basis vectors are linearly dependent, and the system may have infinitely many solutions or no solution.
- ▶ The structure of solutions to a homogeneous system: the number of solutions depends on whether the column vectors are linearly independent.
- ▶ Overdetermined systems: when there are more equations than unknowns, whether a solution exists and the geometric perspective on it.

This section reviews the third perspective on matrix multiplication — a linear combination of the left-hand matrix's column vectors. First, using the chicken-and-rabbit problem, we show that matrix multiplication can be viewed as a linear combination of column vectors, and we review the concept of a basis. We then discuss the characteristics of identity matrices, diagonal matrices, upper triangular matrices, and lower triangular matrices from this perspective, showing how their bases affect vector transformations. Next, we analyze the relationship between the invertibility of a general square matrix and its determinant, showing that when the determinant is zero, the system may have no solution or infinitely many solutions. Finally, we explore the possible solutions of overdetermined systems, analyzing the number of solutions using the third perspective on matrix multiplication.

Reviewing the Third Perspective on Matrix Multiplication: Linear Combination of Column Vectors

This book introduced the third perspective on matrix multiplication earlier; let's briefly review it below.

As shown in Figure 1, for the matrix multiplication $A\mathbf{x} = \mathbf{b}$, we can write the left-hand matrix A as column vectors and expand it to get

$$A_{m \times p} \mathbf{x}_{p \times 1} = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_p \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{bmatrix} = x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 + \cdots + x_p \mathbf{a}_p = \mathbf{b}$$

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This perspective tells us that the column vector $\mathbf{b}$ is a linear combination of matrix $\mathbf{A}$'s column vectors.

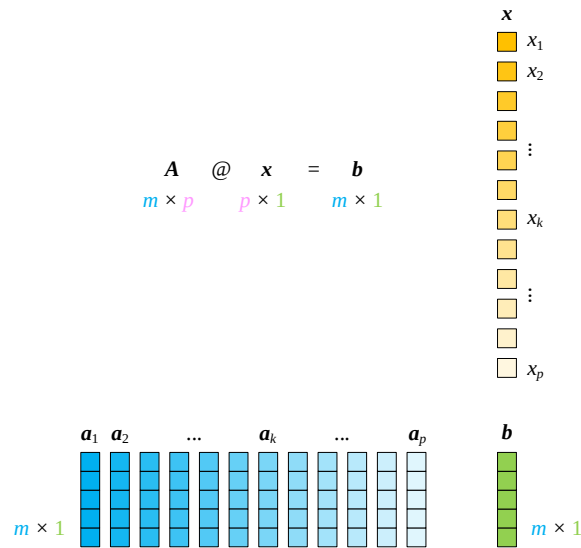


Figure 1. The Third Perspective on the Matrix Multiplication $\mathbf{Ax} = \mathbf{b}$

Revisiting the Chicken-and-Rabbit Problem

In the previous section, we wrote the chicken-and-rabbit problem as the matrix multiplication

$$\mathbf{Ax} = \mathbf{b}$$

where,

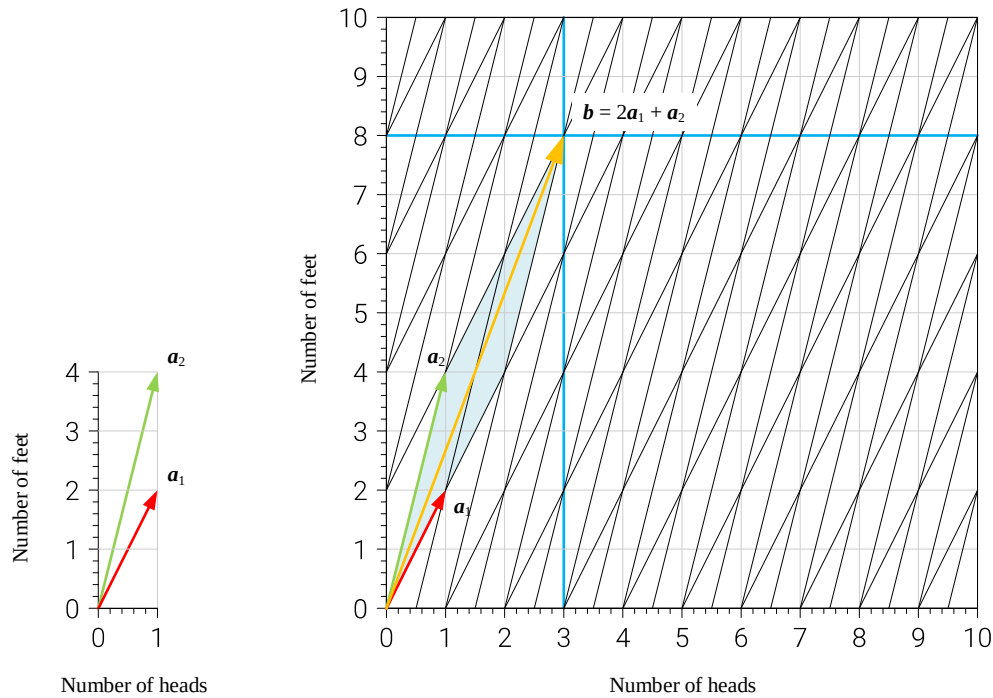
$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 35 \\ 94 \end{bmatrix}$$

Using the third perspective on matrix multiplication to expand

$$\mathbf{Ax} = \begin{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} & \begin{bmatrix} 1 \\ 4 \end{bmatrix} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 35 \\ 94 \end{bmatrix}$$

$\mathbf{a}_1 \quad \mathbf{a}_2 \quad \mathbf{x} \quad \mathbf{b}$

Matrix $\mathbf{A}$'s first column vector $\mathbf{a}_1$ represents one chicken having 1 head and 2 feet; $\mathbf{A}$'s second column vector $\mathbf{a}_2$ represents one rabbit having 1 head and 4 feet.

Figure 2. The Vector $\mathbf{b}$ Is a Linear Combination of $\mathbf{a}_1$ and $\mathbf{a}_2$

Expanding using the third perspective on matrix multiplication gives

$$x_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 35 \\ 94 \end{bmatrix}$$

$\mathbf{a}_1 \quad \mathbf{a}_2 \quad \mathbf{b}$

The vector $\mathbf{b}$ is a linear combination of $\mathbf{a}_1$ and $\mathbf{a}_2$.

Let's take a simpler example: if there are 2 chickens (x_1) and 1 rabbit (x_2), written in linear-combination form,

$$2\mathbf{a}_1 + 1\mathbf{a}_2 = 2 \times \begin{bmatrix} 1 \\ 2 \end{bmatrix} + 1 \times \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 3 \\ 8 \end{bmatrix}$$

$\mathbf{a}_1 \quad \mathbf{a}_2 \quad \mathbf{b}$

As shown in Figure 2, the basis $[\mathbf{a}_1, \mathbf{a}_2]$ spans this plane, and the vector $\mathbf{b}$ is clearly on this plane. And (2, 1) is exactly the vector $\mathbf{b}$'s coordinates in the basis $[\mathbf{a}_1, \mathbf{a}_2]$.

Next, using the linear-combination perspective, let's go back through the examples from the previous section one by one! Then we'll apply the same perspective to analyze overdetermined systems.

Please read this section alongside the previous section.

The Identity Matrix

If the matrix $\mathbf{A}$ is the identity matrix, expanding using the third perspective on matrix multiplication gives

$$\begin{cases} x_1 = 1 \\ x_2 = -1 \end{cases} \Rightarrow \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_x = \underbrace{\begin{bmatrix} 1 \\ -1 \end{bmatrix}}_b \Rightarrow x_1 \underbrace{\begin{bmatrix} 1 \\ 0 \end{bmatrix}}_{a_1} + x_2 \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{a_2} = \underbrace{\begin{bmatrix} 1 \\ -1 \end{bmatrix}}_b$$

As shown in Figure 3, the grid corresponding to the basis $[a_1, a_2]$ is a unit square; that is, a_1 is equivalent to e_1 , and a_2 is equivalent to e_2 .

The vector b can be written as the following linear combination

$$b = a_1 - a_2 = 1 \times \begin{bmatrix} 1 \\ 0 \end{bmatrix} + (-1) \times \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

This shows that the vector b 's coordinates in the basis $[a_1, a_2]$ are $(1, -1)$.

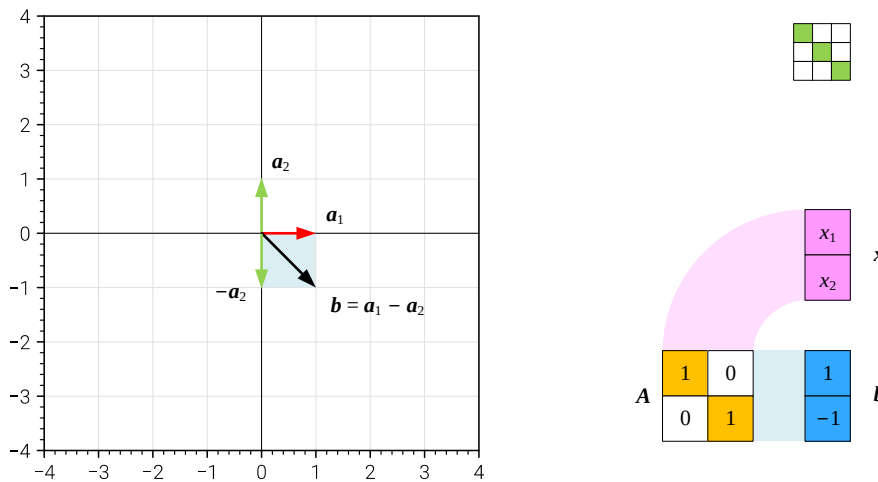


Figure 3. The Matrix A Is the Identity Matrix

Diagonal Matrices: Equal Main-Diagonal Elements

If the matrix A is a diagonal matrix with equal main-diagonal elements, expanding using the third perspective on matrix multiplication gives

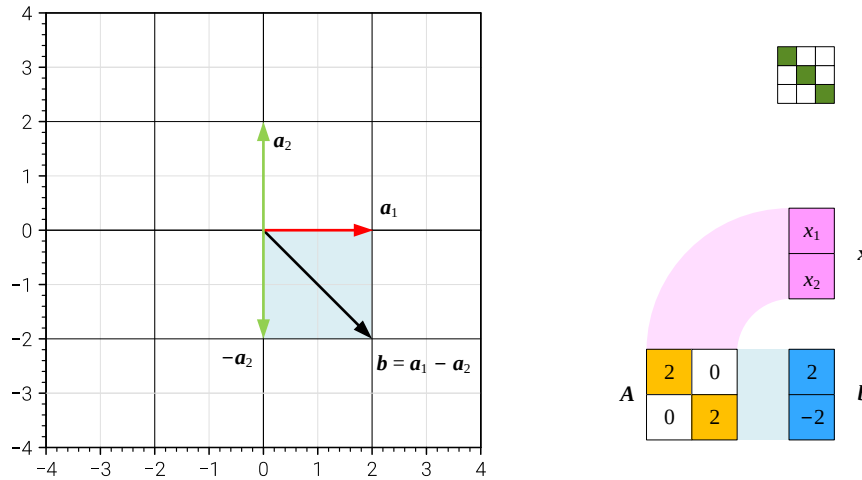
$$\begin{cases} 2x_1 = 2 \\ 2x_2 = -2 \end{cases} \Rightarrow \underbrace{\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_x = \underbrace{\begin{bmatrix} 2 \\ -2 \end{bmatrix}}_b \Rightarrow x_1 \underbrace{\begin{bmatrix} 2 \\ 0 \end{bmatrix}}_{a_1} + x_2 \underbrace{\begin{bmatrix} 0 \\ 2 \end{bmatrix}}_{a_2} = \underbrace{\begin{bmatrix} 2 \\ -2 \end{bmatrix}}_b$$

As shown in Figure 4, the grid corresponding to the basis $[a_1, a_2]$ is a square.

The vector b can be written as the following linear combination

$$b = a_1 - a_2 = \begin{bmatrix} 2 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$$

This means the vector b 's coordinates in the basis $[a_1, a_2]$ are also $(1, -1)$.

Figure 4. The Matrix A Is a Diagonal Matrix with Equal Main-Diagonal Elements

Diagonal Matrices: Unequal Main-Diagonal Elements

If the matrix A is a diagonal matrix with unequal main-diagonal elements, expanding using the third perspective on matrix multiplication gives

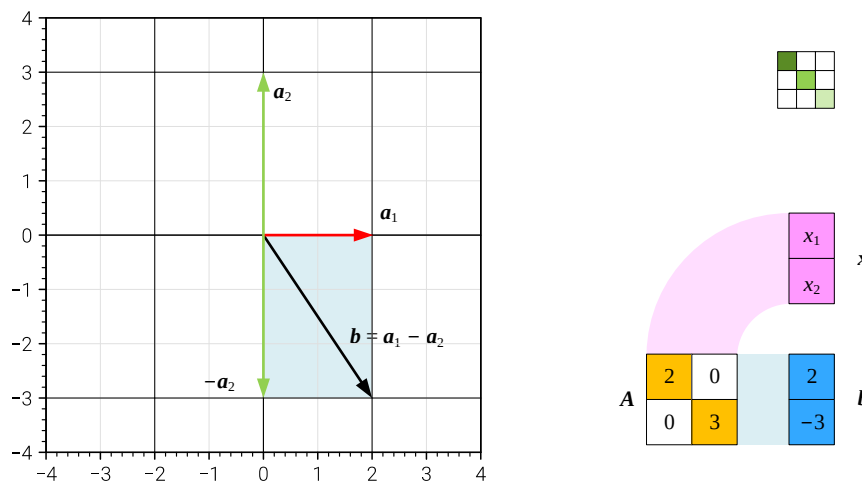
$$\begin{cases} 2x_1 = 2 \\ 3x_2 = -3 \end{cases} \Rightarrow \underbrace{\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_x = \underbrace{\begin{bmatrix} 2 \\ -3 \end{bmatrix}}_b \Rightarrow x_1 \underbrace{\begin{bmatrix} 2 \\ 0 \end{bmatrix}}_{a_1} + x_2 \underbrace{\begin{bmatrix} 0 \\ 3 \end{bmatrix}}_{a_2} = \underbrace{\begin{bmatrix} 2 \\ -3 \end{bmatrix}}_b$$

As shown in Figure 5, the grid corresponding to the basis $[a_1, a_2]$ is a rectangle.

The vector b can be written as the following linear combination

$$b = a_1 - a_2 = \begin{bmatrix} 2 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$$

This shows that the vector b 's coordinates in the basis $[a_1, a_2]$ are also $(1, -1)$.

Figure 5. The Matrix A Is a Diagonal Matrix with Unequal Main-Diagonal Elements

Upper Triangular Matrices

If the matrix A is an upper triangular matrix, expanding using the third perspective on matrix multiplication gives

$$\begin{cases} x_1 + x_2 = 0 \\ x_2 = -1 \end{cases} \Rightarrow \underbrace{\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_x = \underbrace{\begin{bmatrix} 0 \\ -1 \end{bmatrix}}_b \Rightarrow x_1 \underbrace{\begin{bmatrix} 1 \\ 0 \end{bmatrix}}_{a_1} + x_2 \underbrace{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}_{a_2} = \underbrace{\begin{bmatrix} 0 \\ -1 \end{bmatrix}}_b$$

As shown in Figure 6, the grid corresponding to the basis $[a_1, a_2]$ is a parallelogram (the unit square sheared along the horizontal axis).

The vector b can be written as the following linear combination

$$b = a_1 - a_2 = \begin{bmatrix} 2 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$$

Please work out for yourself the vector b 's coordinates in the basis $[a_1, a_2]$.

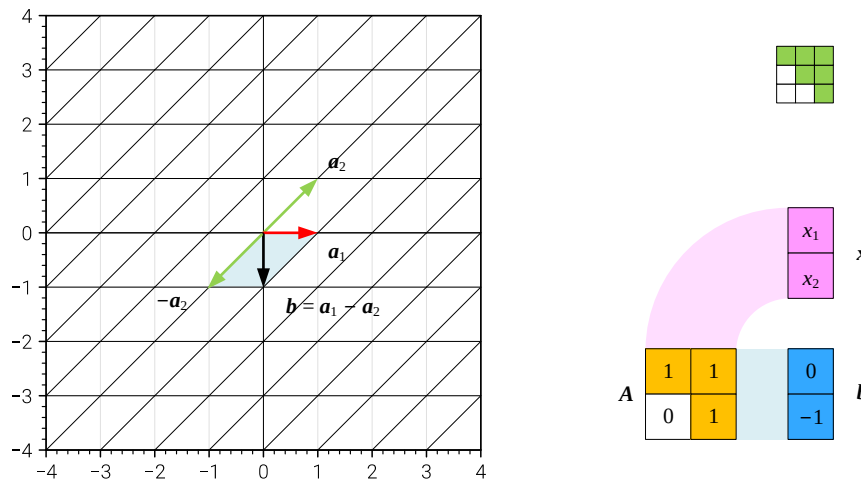


Figure 6. The Matrix A Is an Upper Triangular Matrix

Lower Triangular Matrices

If the matrix A is a lower triangular matrix, expanding using the third perspective on matrix multiplication gives

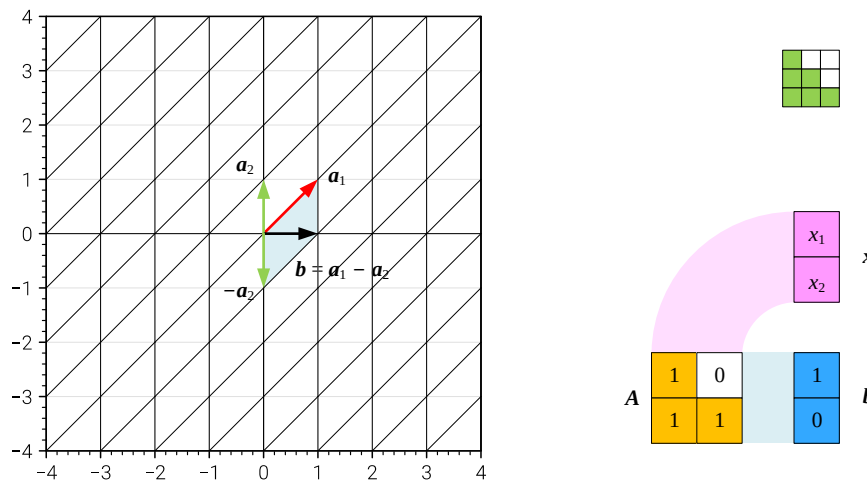
$$\begin{cases} x_1 = 1 \\ x_1 + x_2 = 0 \end{cases} \Rightarrow \underbrace{\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_x = \underbrace{\begin{bmatrix} 1 \\ 0 \end{bmatrix}}_b \Rightarrow x_1 \underbrace{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}_{a_1} + x_2 \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{a_2} = \underbrace{\begin{bmatrix} 1 \\ 0 \end{bmatrix}}_b$$

As shown in Figure 7, the grid corresponding to the basis $[a_1, a_2]$ is a parallelogram (the unit square sheared along the vertical axis).

The vector b can be written as the following linear combination

$$b = a_1 - a_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Please work out for yourself the vector b 's coordinates in the basis $[a_1, a_2]$.

Figure 7. The Matrix A Is a Lower Triangular Matrix

General Square Matrices with Nonzero Determinant

If the matrix A is a general square matrix, and its determinant is nonzero, then A is invertible and the system has a unique solution — for example, the following.

Expanding using the third perspective on matrix multiplication gives

$$\begin{cases} x_1 - x_2 = 2 \\ 2x_1 + 2x_2 = 0 \end{cases} \Rightarrow \underbrace{\begin{bmatrix} 1 & -1 \\ 2 & 2 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_x = \underbrace{\begin{bmatrix} 2 \\ 0 \end{bmatrix}}_b \Rightarrow x_1 \underbrace{\begin{bmatrix} 1 \\ 2 \end{bmatrix}}_{a_1} + x_2 \underbrace{\begin{bmatrix} -1 \\ 2 \end{bmatrix}}_{a_2} = \underbrace{\begin{bmatrix} 2 \\ 0 \end{bmatrix}}_b$$

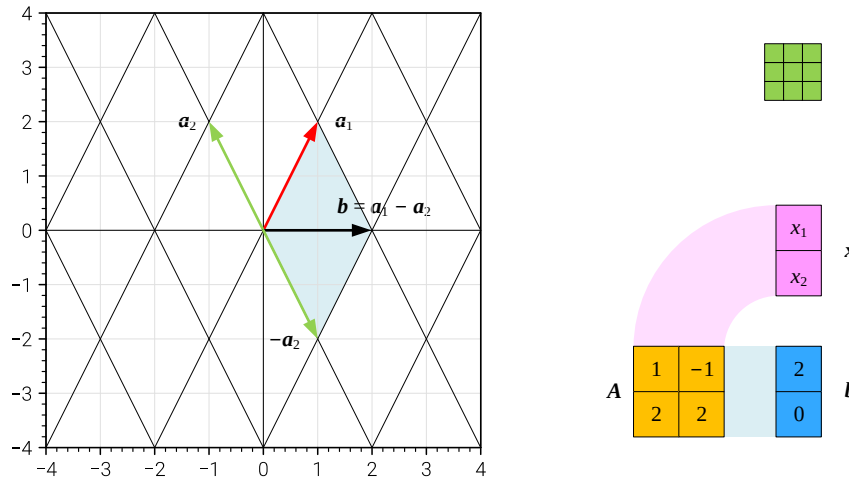
As shown in Figure 8, the grid corresponding to the basis $[a_1, a_2]$ is a parallelogram.

The vector b can be written as the following linear combination

$$b = a_1 - a_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} - \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$



Please work out for yourself the vector b 's coordinates in the basis $[a_1, a_2]$.

Figure 8. The Matrix A Has Nonzero Determinant, $Ax = b$

Homogeneous Linear Systems: Unique Solution

If a system of linear equations is homogeneous, that is, $b = 0$ – for example, the following – expanding using the third perspective on matrix multiplication gives

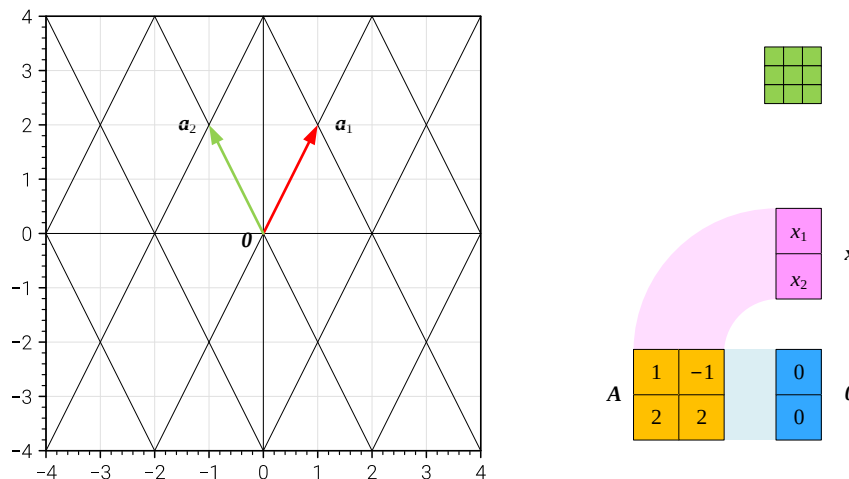
$$\begin{cases} x_1 - x_2 = 0 \\ 2x_1 + 2x_2 = 0 \end{cases} \Rightarrow \underbrace{\begin{bmatrix} 1 & -1 \\ 2 & 2 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_x = \underbrace{\begin{bmatrix} 0 \\ 0 \end{bmatrix}}_0 \Rightarrow x_1 \underbrace{\begin{bmatrix} 1 \\ 2 \end{bmatrix}}_{a_1} + x_2 \underbrace{\begin{bmatrix} -1 \\ 2 \end{bmatrix}}_{a_2} = \underbrace{\begin{bmatrix} 0 \\ 0 \end{bmatrix}}_0$$

As shown in Figure 9, the grid corresponding to the basis $[a_1, a_2]$ is a parallelogram, exactly the same as in Figure 8.

The vector b can be written as the following linear combination

$$0 = a_1 - a_2 = 0 \times \begin{bmatrix} 1 \\ 2 \end{bmatrix} + 0 \times \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Since a_1 and a_2 are linearly independent, the zero vector 0 's coordinates in the basis $[a_1, a_2]$ are $(0, 0)$, which is the unique solution to the homogeneous system.

Figure 9. The Matrix A Has Nonzero Determinant, $Ax = 0$

General Square Matrices with Zero Determinant

Infinitely Many Solutions

In the example below, the determinant of the matrix A is 0.

$$\begin{cases} x_1 + x_2 = -1 \\ 2x_1 + 2x_2 = -2 \end{cases} \Rightarrow \underbrace{\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_x = \underbrace{\begin{bmatrix} -1 \\ -2 \end{bmatrix}}_b \Rightarrow x_1 \underbrace{\begin{bmatrix} 1 \\ 2 \end{bmatrix}}_{a_1} + x_2 \underbrace{\begin{bmatrix} 1 \\ 2 \end{bmatrix}}_{a_2} = \underbrace{\begin{bmatrix} -1 \\ -2 \end{bmatrix}}_b$$

As shown in Figure 10, a_1 and a_2 are linearly dependent, so clearly $[a_1, a_2]$ cannot form a basis.

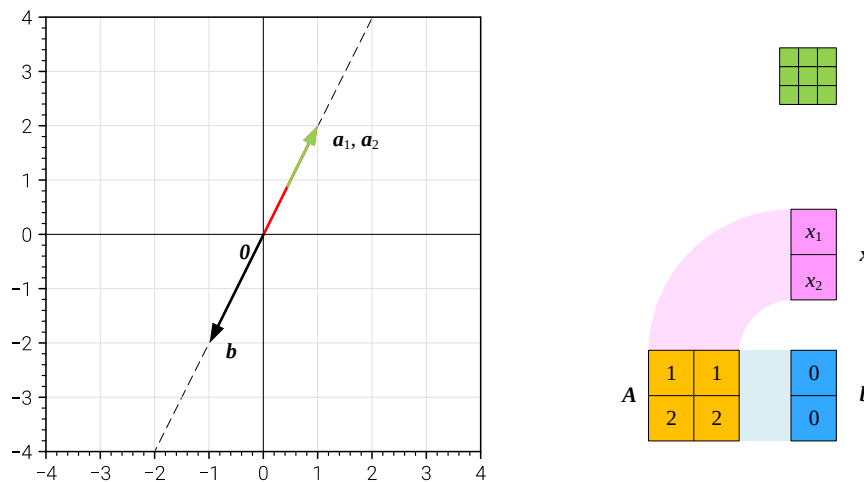


Figure 10. The Matrix A Has Zero Determinant, $Ax = b$ – Infinitely Many Solutions

But the equation above does have solutions, because b and a_1, a_2 lie on the same line. The values of x_1 and x_2 satisfying the equation below are the solutions of the system

$$k \underbrace{\begin{bmatrix} 1 \\ 2 \end{bmatrix}}_{a_1} - (k+1) \underbrace{\begin{bmatrix} 1 \\ 2 \end{bmatrix}}_{a_2} = \underbrace{\begin{bmatrix} -1 \\ -2 \end{bmatrix}}_b$$

where k is any real number. This means that the homogeneous linear system in has infinitely many solutions, that is,

$$\begin{cases} x_1 = k \\ x_2 = -k - 1 \end{cases}$$

No Solution

As with the previous example, in the example below the determinant of the matrix A is also 0; expanding using the third perspective on matrix multiplication gives

$$\begin{cases} x_1 + x_2 = 0 \\ 2x_1 + 2x_2 = -2 \end{cases} \Rightarrow \underbrace{\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_x = \underbrace{\begin{bmatrix} 0 \\ -2 \end{bmatrix}}_b \Rightarrow x_1 \underbrace{\begin{bmatrix} 1 \\ 2 \end{bmatrix}}_{a_1} + x_2 \underbrace{\begin{bmatrix} 1 \\ 2 \end{bmatrix}}_{a_2} = \underbrace{\begin{bmatrix} 0 \\ -2 \end{bmatrix}}_b$$

As shown in Figure 11, the vectors $\mathbf{a}_1$ and $\mathbf{a}_2$ coincide, and their linear combinations all lie on the line drawn in the figure. Clearly, no matter how $\mathbf{a}_1$ and $\mathbf{a}_2$ are linearly combined, the vector $\mathbf{b}$ can never lie on this line.

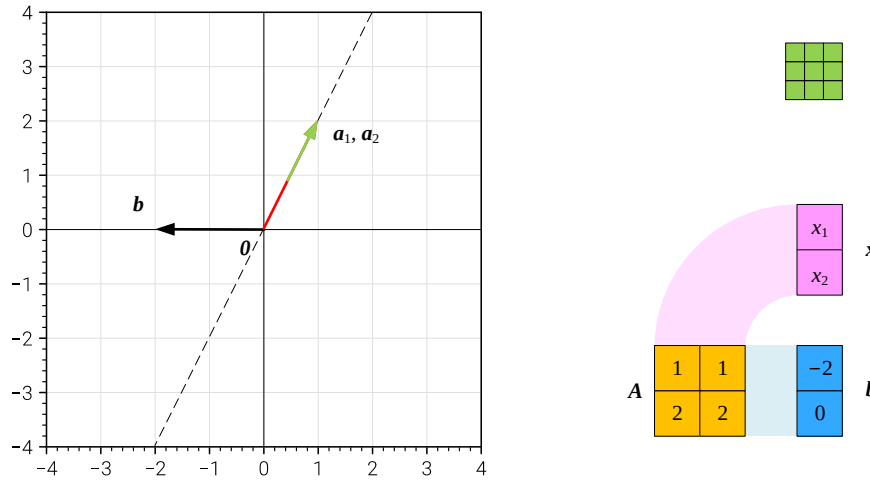


Figure 11. The Matrix $\mathbf{A}$ Has Zero Determinant, $\mathbf{Ax} = \mathbf{b}$ — No Solution

Homogeneous Linear Systems: Infinitely Many Solutions

Next, let's look at another example of a homogeneous linear system.

For the following system of linear equations, $\det(\mathbf{A}) = 0$; let's first write it as matrix multiplication, and then expand it using the third perspective on matrix multiplication

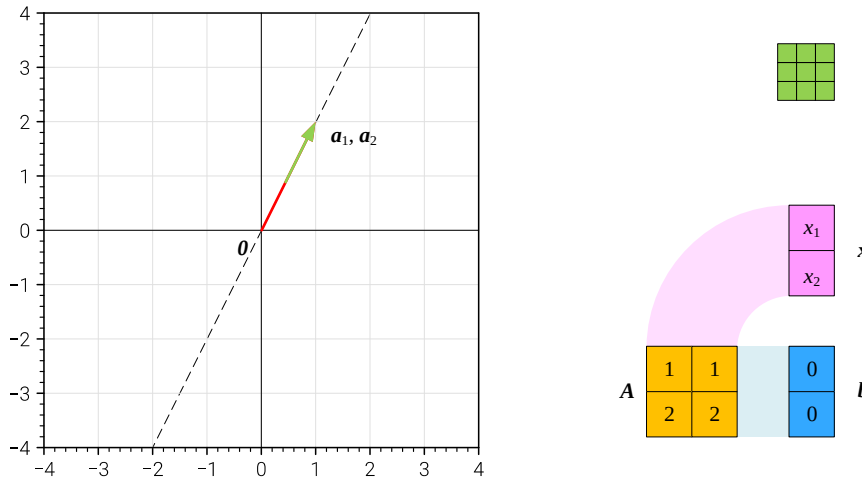
$$\begin{cases} x_1 + x_2 = 0 \\ 2x_1 + 2x_2 = 0 \end{cases} \Rightarrow \underbrace{\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_{\mathbf{x}} = \underbrace{\begin{bmatrix} 0 \\ 0 \end{bmatrix}}_{\mathbf{b}} \Rightarrow x_1 \underbrace{\begin{bmatrix} 1 \\ 2 \end{bmatrix}}_{\mathbf{a}_1} + x_2 \underbrace{\begin{bmatrix} 1 \\ 2 \end{bmatrix}}_{\mathbf{a}_2} = \underbrace{\begin{bmatrix} 0 \\ 0 \end{bmatrix}}_{\mathbf{b}}$$

As shown in Figure 12, the vectors $\mathbf{a}_1$ and $\mathbf{a}_2$ coincide; the vectors $\mathbf{a}_1$ and $\mathbf{a}_2$'s linear combinations also lie on Figure 12's line through the origin; the zero vector $\mathbf{0}$ is clearly on this line as well.

For any given real number k , the zero vector can be obtained through the following linear combination $\mathbf{0}$

$$k \underbrace{\begin{bmatrix} 1 \\ 2 \end{bmatrix}}_{\mathbf{a}_1} - k \underbrace{\begin{bmatrix} 1 \\ 2 \end{bmatrix}}_{\mathbf{a}_2} = \underbrace{\begin{bmatrix} 0 \\ 0 \end{bmatrix}}_{\mathbf{b}}$$

This shows that $(k, -k)$ is a solution of the system for every k ; clearly, this homogeneous linear system has infinitely many solutions.

Figure 12. The Matrix A Has Zero Determinant, $Ax = 0$ – Infinitely Many Solutions

Overdetermined Systems

An overdetermined system is a system of equations in which the number of equations exceeds the number of unknowns.

For example, a system made up of three linear equations in two variables has the following basic form:

$$\begin{cases} a_{1,1}x_1 + a_{1,2}x_2 = b_1 \\ a_{2,1}x_1 + a_{2,2}x_2 = b_2 \\ a_{3,1}x_1 + a_{3,2}x_2 = b_3 \end{cases}$$

The three linear equations above can be written in matrix form:

$$\underbrace{\begin{bmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \\ a_{3,1} & a_{3,2} \end{bmatrix}}_A \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_x = \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}}_b$$

The coefficient matrix is a 3×2 matrix

$$A_{3 \times 2} = \begin{bmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \\ a_{3,1} & a_{3,2} \end{bmatrix}$$

The variable vector x is two-dimensional

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

The constant vector b is three-dimensional, corresponding to the number of equations

$$b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

Since this is an overdetermined system, we need to analyze the possible types of solutions it can have. Geometrically, each equation represents a line, so the type of solution depends on how the three lines relate to each other in the plane.

Overdetermined Linear Systems: No Solution

Figure 13 shows four systems, all of which have no solution.

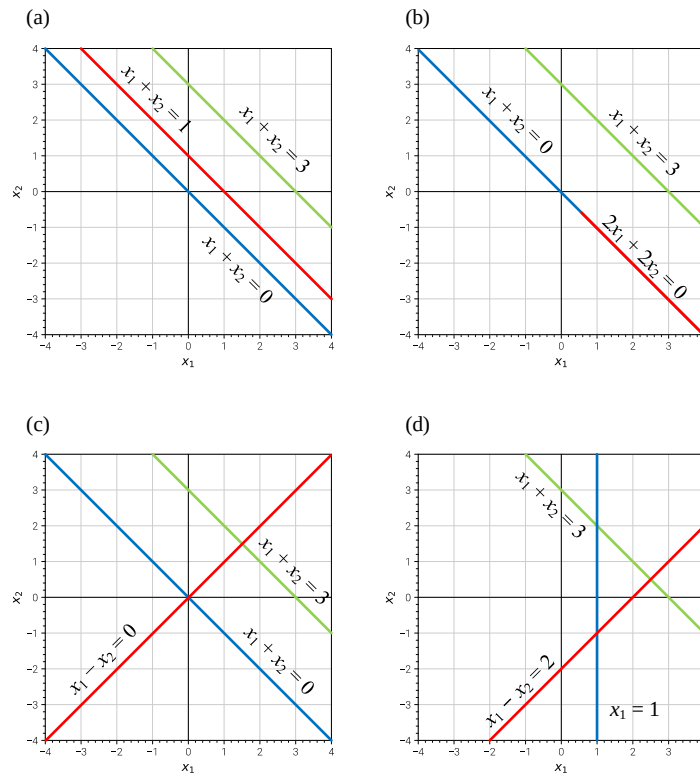


Figure 13. Overdetermined Systems (Non-Homogeneous), No Solution

For example, in Figure 13 (a), the three lines are all parallel to each other and share no common intersection, so the system has no solution.

Writing the system in matrix-multiplication form

$$\begin{cases} x_1 + x_2 = 1 \\ x_1 + x_2 = 3 \\ x_1 + x_2 = 0 \end{cases} \Rightarrow \underbrace{\begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_x = \underbrace{\begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}}_b$$

Expanding the matrix multiplication above using the third perspective on matrix multiplication gives

$$x_1 \underbrace{\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}}_{a_1} + x_2 \underbrace{\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}}_{a_2} = \underbrace{\begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}}_b$$

As shown in Figure 14, clearly $\mathbf{a}_1$ and $\mathbf{a}_2$ are collinear, while the vector $\mathbf{b}$ is not on this line (through the origin); so this system of linear equations has no solution.

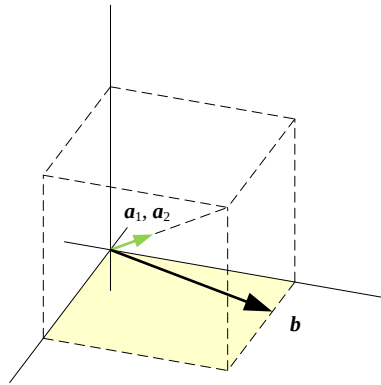


Figure 14. $\mathbf{a}_1$ and $\mathbf{a}_2$ Are Collinear, While the Vector $\mathbf{b}$ Is Not on This Line (Through the Origin)

Please use the same method to analyze the remaining three systems of linear equations in Figure 13.

Overdetermined Linear Systems: Unique Solution

Figure 15 shows two cases where an overdetermined system has a unique solution.

Figure 15 (a): two of the lines coincide, and they intersect the third line at one point.

Figure 15 (b): the three lines intersect at a single point, and this intersection point is the unique solution of the overdetermined linear system.

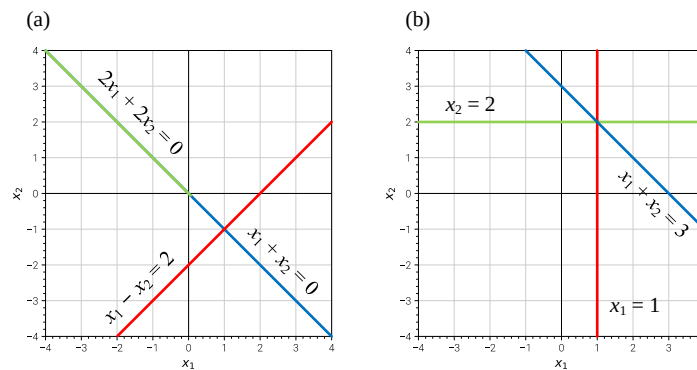


Figure 15. Overdetermined Systems (Non-Homogeneous), Unique Solution

Writing the system of equations corresponding to Figure 15 (b) in matrix-multiplication form

$$\begin{cases} x_1 = 1 \\ x_2 = 2 \\ x_1 + x_2 = 3 \end{cases} \Rightarrow \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_x = \underbrace{\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}}_b$$

Expanding the matrix multiplication above using the third perspective on matrix multiplication gives

$$1 \times \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + 2 \times \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$\mathbf{a}_1 \qquad \mathbf{a}_2 \qquad \mathbf{b}$

As shown in Figure 16, clearly $\mathbf{a}_1$ and $\mathbf{a}_2$ are not collinear, but $\mathbf{a}_1$, $\mathbf{a}_2$, and $\mathbf{b}$ are coplanar; the vector $\mathbf{b}$ can be written as a linear combination of $\mathbf{a}_1$ and $\mathbf{a}_2$.

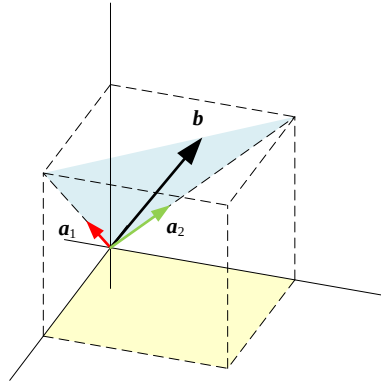


Figure 16. $\mathbf{a}_1$ and $\mathbf{a}_2$ Are Not Collinear, but $\mathbf{a}_1$, $\mathbf{a}_2$, and $\mathbf{b}$ Are Coplanar

Figure 17 shows two homogeneous overdetermined systems, each with a unique solution located at the origin.

 Please analyze these two figures in Figure 17 on your own.

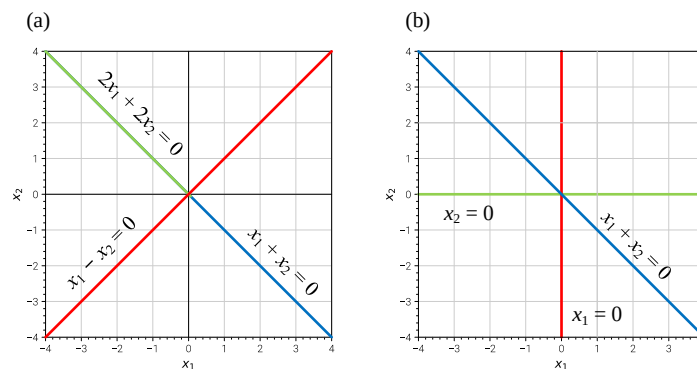


Figure 17. Overdetermined Systems (Homogeneous), Unique Solution

Overdetermined Linear Systems: Infinitely Many Solutions

As shown in Figure 18 (a), if all three lines coincide – meaning they represent different equation forms of the same line – then the system has infinitely many solutions.

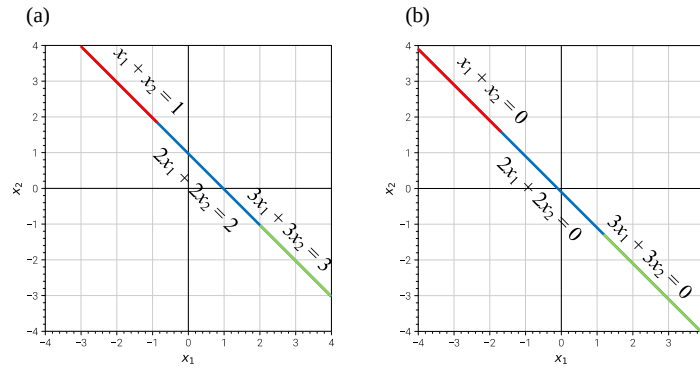


Figure 18. Overdetermined Systems, Infinitely Many Solutions

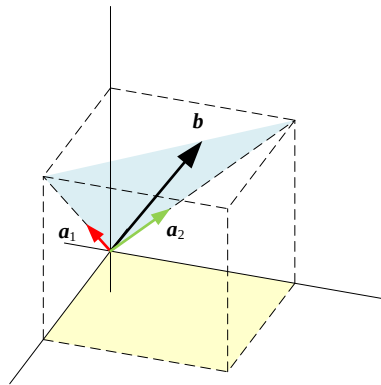
Writing the system of equations corresponding to Figure 18 (a) in matrix-multiplication form

$$\begin{cases} x_1 + x_2 = 1 \\ 2x_1 + 2x_2 = 2 \\ 3x_1 + 3x_2 = 3 \end{cases} \Rightarrow \underbrace{\begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 3 & 3 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_x = \underbrace{\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}}_b$$

Expanding the matrix multiplication above using the third perspective on matrix multiplication gives

$$(k+1) \times \underbrace{\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}}_{a_1} - k \times \underbrace{\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}}_{a_2} = \underbrace{\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}}_b$$

As shown in Figure 19, a_1 , a_2 , and b are collinear; more precisely, all three coincide.

Figure 19. a_1 , a_2 , and b Are Collinear

For the linear combination above, k can be any real number; the linear combination $(k+1)a_1$ and $-ka_2$ gives b .

Figure 18 (b)'s system of equations is a homogeneous linear system. As mentioned in the previous section, a homogeneous linear system is one where b is the zero vector 0 .

Writing the system of equations corresponding to Figure 18 (b) in matrix-multiplication form

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$$\begin{cases} x_1 + x_2 = 0 \\ 2x_1 + 2x_2 = 0 \\ 3x_1 + 3x_2 = 0 \end{cases} \Rightarrow \underbrace{\begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 3 & 3 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_x = \underbrace{\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}}_{b=0}$$

Expanding the matrix multiplication above using the third perspective on matrix multiplication gives

$$k \times \underbrace{\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}}_{a_1} + (-k) \times \underbrace{\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}}_{a_2} = \underbrace{\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}}_{b=0}$$

For the linear combination above, k can be any real number; the linear combination $k\mathbf{a}_1$ and $(-k)\mathbf{a}_2$ gives $\mathbf{0}$. Therefore, has infinitely many solutions.



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. Please write the following systems of linear equations in matrix-multiplication form.

▶ $\begin{cases} x_1 + x_2 = 3 \\ 2x_1 + 4x_2 = 10 \end{cases}$

▶ $\begin{cases} x_1 + x_2 = 3 \\ x_1 + 2x_2 = 5 \end{cases}$

▶ $\begin{cases} x_1 + x_2 = 3 \\ x_2 = 2 \end{cases}$

▶ $\begin{cases} x_1 = 1 \\ x_1 + 2x_2 = 5 \end{cases}$

▶ $\begin{cases} x_1 = 1 \\ x_2 = 2 \end{cases}$

Q2. Please expand the matrix multiplication from the previous question using the third perspective (linear combination of column vectors), and draw the grid.